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Structural Portfolio

October 2026

About

I'm a second-year Civil Engineering student at Toronto Metropolitan University (BEng, expected 2029), focused on structural engineering. I've worked on site with a design-build firm, tying rebar, building formwork and placing concrete, and as a project coordinator on a 182-unit mixed-use development, working between the structural, civil, geotechnical, planning and landscape consultants.

Outside class I build verification packages: I check a structure by hand, in a spreadsheet and in analysis software, chase down every difference between them, and issue drawings and models under revision control. My first package was reviewed by a practising P.Eng. Tools: Robot Structural Analysis, Mathcad Prime, Excel, AutoCAD and Revit (Autodesk Certified User).

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All projects are self-directed student exercises and are not for construction.

Project 1 – Maintenance Platform: Structural Verification

Self-directed project | October 2026 | Reviewed by S. Azoubaidy, P.Eng. | R0 issued for P.Eng. review; R1 in progress

Tools: Mathcad Prime, Excel, Robot Structural Analysis, AutoCAD, Revit

3	independent verification methods (Mathcad, Excel, Robot)
≤1.7%	maximum difference between methods (C1 post load)
6.0 × 3.0 m	steel platform, top of steel 3,000 mm
1,298 kg	steel take-off (CISC mass per metre, members only)

Description

A 6.0 m × 3.0 m steel maintenance platform (W310 beams, W150 joists, HSS posts, L-angle X-bracing), checked for gravity loads by three independent methods: hand calculation in Mathcad, a parametric Excel workbook and a 3D Robot model. Drawn in AutoCAD and modelled in Revit, with a material take-off and a review register. Issued for review to S. Azoubaidy, P.Eng.; revision R1 (tension-only bracing and guard lateral loads) in progress. Student exercise: elastic analysis only, not a CSA S16 design.

Key highlights

- Beam and joist results agreed within 0.1% across Mathcad, Excel and Robot.
- Traced a 1.7% post-load difference to bracing self-weight, and found the X-bracing carrying about 4.9 kN of gravity load.
- Found the long braces over the slenderness limit (L/r ≈ 537 vs 300), so specified a bolted crossing and tension-only bracing.
- Caught a modelling error from the Revit schedule (braces modelled as W-sections) and corrected it. The take-off matched the Revit volumes within 2.3%.

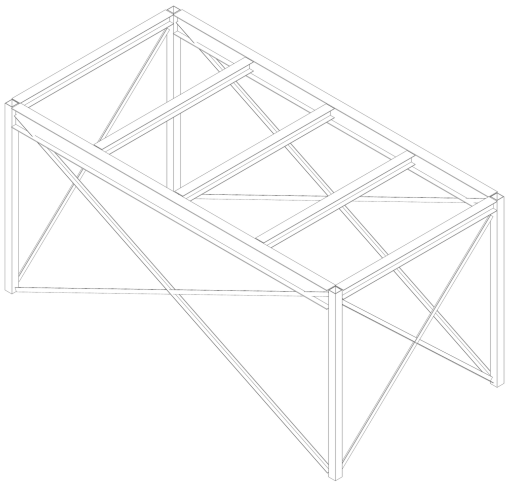


Figure 1: Revit 3D model (AS-P1-MOD-002).

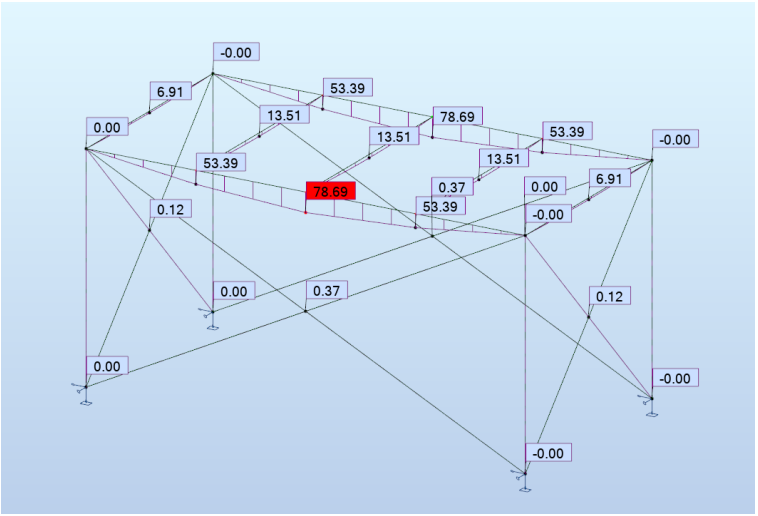


Figure 2: Robot ULS bending moments (kN·m); B1 maximum 78.69 kN·m.

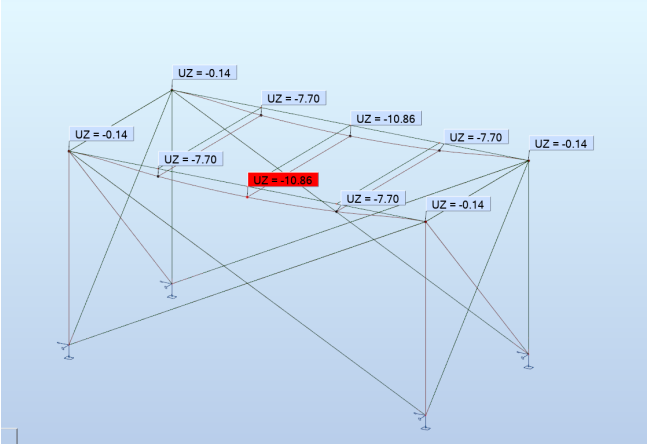


Figure 3: Robot SLS vertical displacement UZ (mm).

P1 Steel Framing Schedule

Mark	Type	Structural Usage	Cut Length	Count	Volume
BR1-1	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-2	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-3	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-4	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-5	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-6	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-7	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-8	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
L2-1/2X2-1/2X1/4: 8			41243 mm		0.0318 m³
J1-1	W6X15	Joist	2848 mm	1	0.0084 m³
J1-2	W6X15	Joist	2810 mm	1	0.0083 m³
J1-3	W6X15	Joist	2810 mm	1	0.0083 m³
J1-4	W6X15	Joist	2810 mm	1	0.0083 m³
J1-5	W6X15	Joist	2848 mm	1	0.0084 m³
W6X15: 5			14124 mm		0.0417 m³
B1-1	W12X26	Girder	5848 mm	1	0.0301 m³
B1-2	W12X26	Girder	5848 mm	1	0.0301 m³
W12X26: 2			11695 mm		0.0601 m³
Grand total: 15			67063 mm		0.1336 m³

P1 Column Schedule

ss	Type	Base Level	Top Level	Count	System Length	Volume
C1-1	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-2	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-3	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-4	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
Grand total: 4						0.0355 m³

Figure 4: S-301 Revit steel framing and column schedules.

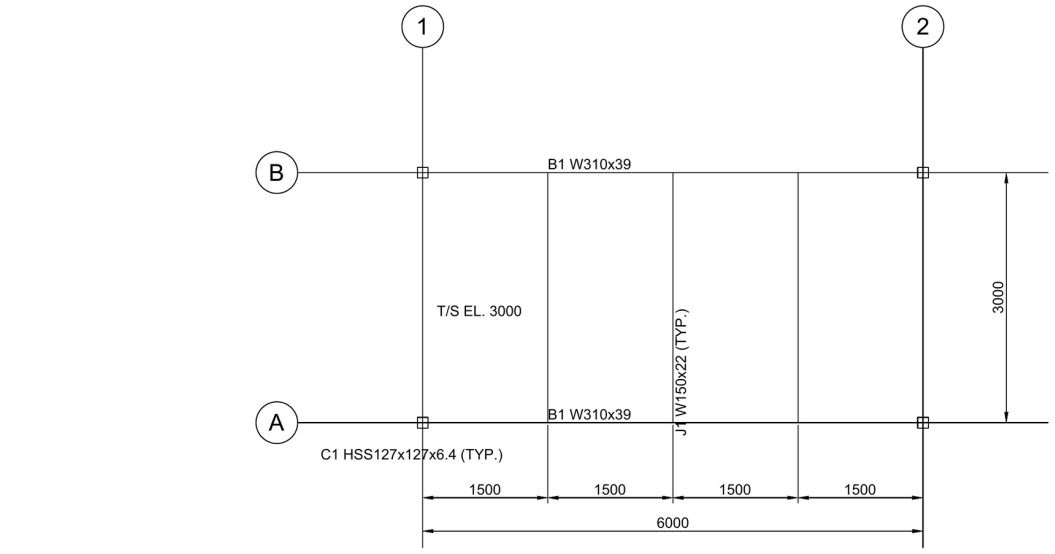
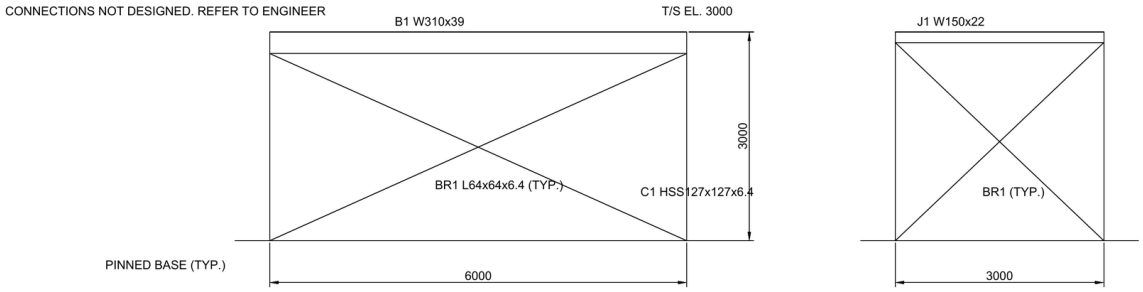


Figure 5: S-101 framing plan, 1:50 (R2).



SECTION 1 - GRID A (GRID B SIMILAR) 1:50

SECTION 2 - GRID 1 (GRID 2 SIMILAR) 1:50

GENERAL NOTES

1. STUDENT EXERCISE. NOT FOR CONSTRUCTION.
2. DESIGN BASIS: ELASTIC STRENGTH OF MATERIALS CHECK ONLY. NO CSA S16 DESIGN CHECK.
3. LOADS: GRATING 0.5 kPa + SELF-WEIGHT (DEAD); LIVE 4.8 kPa; EQUIPMENT 10 kN AT MIDSPAN OF EACH B1. ULS 1.25D + 1.5L.
4. MATERIALS: CSA G40.21 350W (ASSUMED).
5. CONNECTIONS, BRACING AND GUARD LATERAL LOAD NOT DESIGNED. REFER TO ENGINEER.
6. DIMENSIONS IN mm. ELEVATIONS IN mm ABOVE GRADE.
7. SEE CALCULATION AS-P1-CAL-001 AND ROBOT MODEL AS-P1-MOD-001.
8. BR1 BOLTED AT X-BRACE INTERSECTION. BRACING TENSION-ONLY (SLENDERNESS). SEE AS-P1-MTO-001.

Figure 6: S-201 sections and general notes, 1:50 (R2).

Three-way comparison (AS-P1-RPT-001 R0)

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment, M_f	kN·m	13.51	13.52	13.51	0.0%	–
J1 factored shear, V_f	kN	18.02	18.02	18.02	0.0%	–
J1 service deflection	mm	3.56	3.56	3.56	0.0%	8.33
B1 factored moment, M_f	kN·m	78.70	78.70	78.69	0.0%	–
B1 factored shear, V_f	kN	35.96	35.96	35.95	0.0%	–
B1 service deflection	mm	10.72	10.72	10.72	0.0%	16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	–
C1 Euler buckling load, P_{cr}	kN	1,546	1,546	Not run	–	–

C1: hand 46.03 kN vs Robot reaction 46.83 kN, a difference of +0.80 kN (1.7%) per post, from the bracing self-weight (about 2.6 kN in total) that only the Robot model includes. Separately, about 4.9 kN reaches the support through the X-bracing in Robot (reaction 46.83 kN vs post member force 41.9 kN). Comparison criterion: 2%.

Design basis

Members	B1 W310×39 (6.0 m, simply supported); J1 W150×22 at 1.5 m (3.0 m); C1 HSS127×127×6.4 (3.0 m, pinned, K = 1.0); BR1 L64×64×6.4 X-bracing, all four sides
Loads	Dead 0.5 kPa grating + self-weight; Live 4.8 kPa (assumed); Equipment 10 kN at midspan of each B1 (assumed)
Combinations	ULS 1.25D + 1.5L; SLS D + L; Deflection L/360 (assumed)
Material	CSA G40.21 350W (assumed); E = 200,000 MPa
Scope	Elastic analysis only. CSA S16 resistances, lateral-torsional buckling, connections and guard lateral load not yet checked.

Full documents: Appendix A, from page 12.

Project 2 – Conveyor Gallery Truss: Member Force Verification

Self-directed project | October 2026 | Rev 0, first issue

Tools: hand calculation, Robot Structural Analysis, Excel, AutoCAD

24 m	span, 8 panels × 3.0 m, 3.0 m deep
146.4 kN	maximum chord force (U3–U4, compression)
0.0%	hand vs Robot difference on the 7 hand-solved members
1,317 kg	steel take-off, one truss (1,449 kg with 10% connection allowance)

Description

Member-force verification of a 24 m, statically determinate Pratt truss for a conveyor gallery, with 8 panels at 3 m, 3 m deep, and pin and roller supports. Loads were worked out by hand to give reactions of 73.2 kN. Seven members were then solved by method of joints (L0, U0, L1) and a section cut through panel 4. The results were compared against a Robot Structural Analysis model after converting Robot's sign convention (+ = compression). SLS deflection was checked, the elevation was drawn in AutoCAD (S-301) and a material take-off was prepared.

Key highlights

- All 7 hand-solved members match Robot within 0.0%. The remaining 26 were taken from Robot and checked by symmetry.
- Governing chord force U3–U4 = 146.4 kN compression, from the section cut M/d = 439.2 / 3.0. It matches Robot exactly.
- Robot's default self-weight added 12.9 kN to the governing chord force. It was removed so the model matched the hand-calculation load basis.
- SLS midspan deflection is 15.8 mm against the L/360 limit of 66.7 mm (about L/1520). Steel take-off is 1,317 kg per truss, or 1,449 kg with a 10% connection allowance.

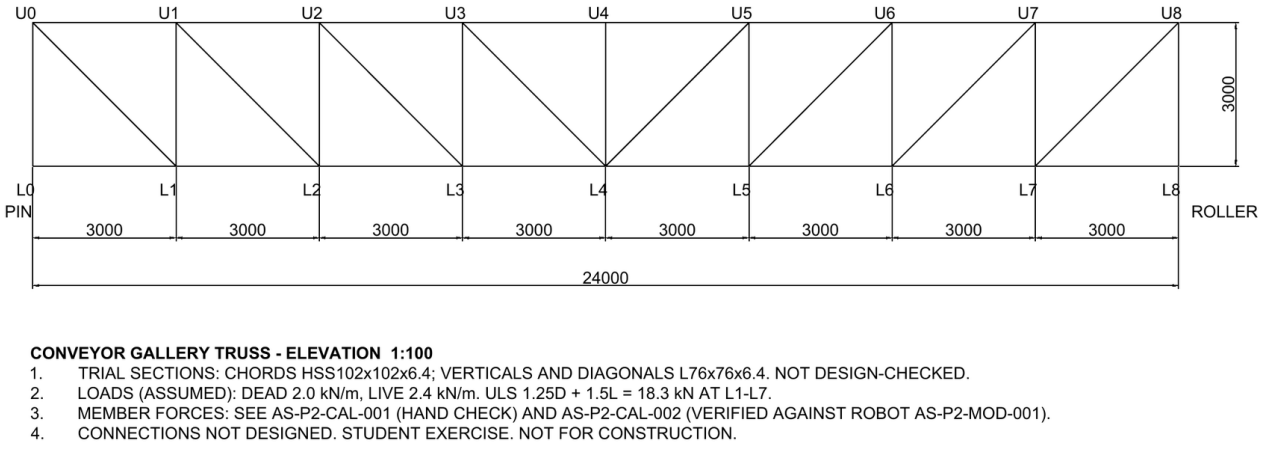


Figure 7: S-301 truss elevation, 1:100 (R0).

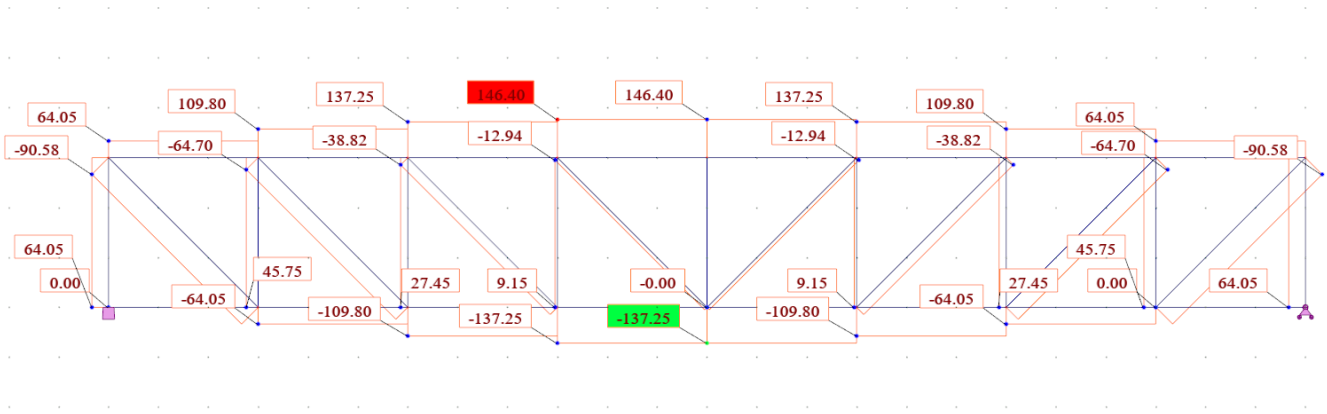


Figure 8: Robot ULS axial forces (kN); maximum 146.40 kN in U3–U4.

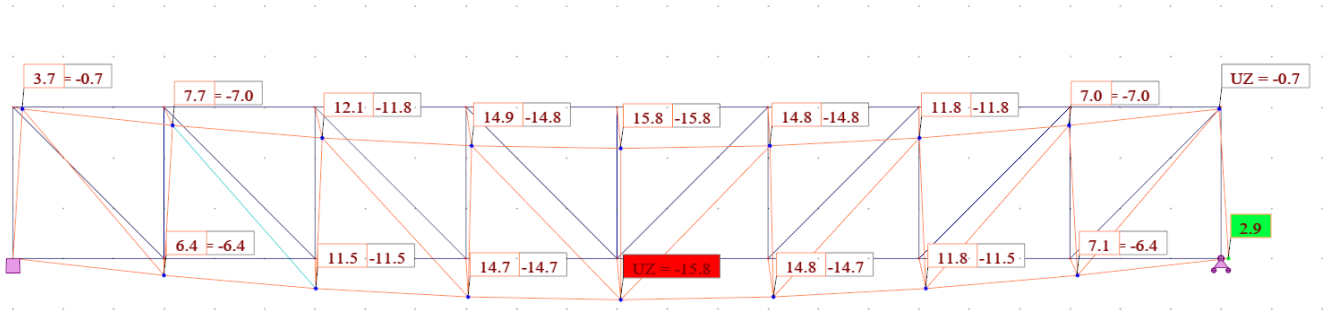


Figure 9: Robot SLS deflected shape, UZ (mm); maximum 15.8 mm at midspan.

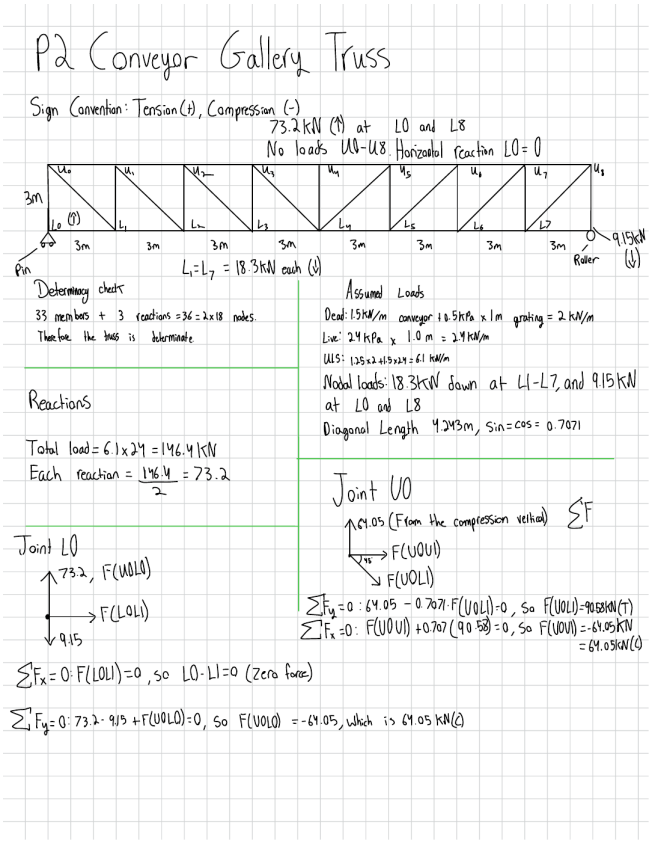
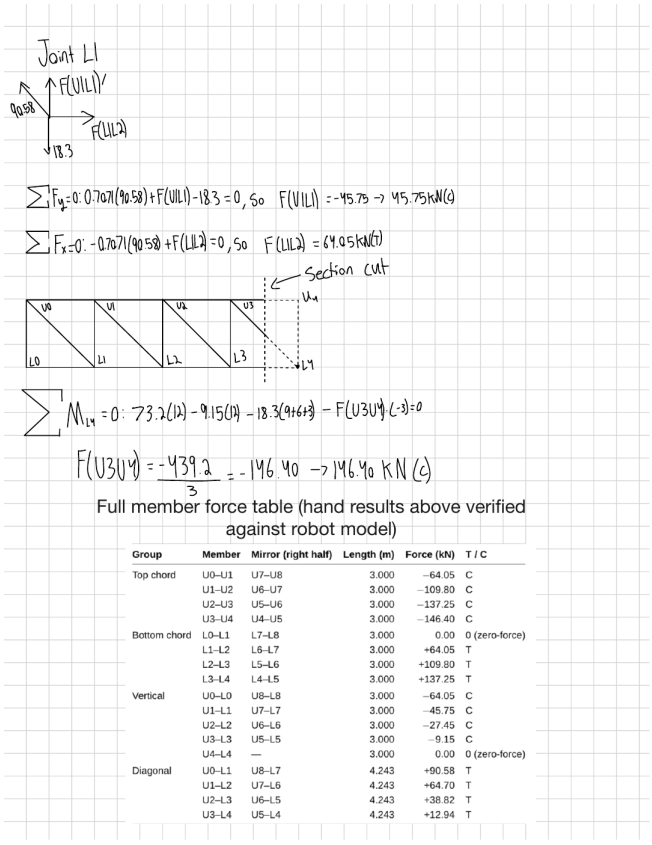


Figure 10: Hand calculation: loads, reactions, joints L0 and U0.



Full member force table (hand results above verified against robot model)

Group	Member	Mirror (right half)	Length (m)	Force (kN)	T / C
Top chord	U0–U1	U7–U8	3.000	−64.05	C
	U1–U2	U6–U7	3.000	−109.80	C
	U2–U3	U5–U6	3.000	−137.25	C
	U3–U4	U4–U5	3.000	−146.40	C
Bottom chord	L0–L1	L7–L8	3.000	0.00	0 (zero-force)
	L1–L2	L6–L7	3.000	+64.05	T
	L2–L3	L5–L6	3.000	+109.80	T
	L3–L4	L4–L5	3.000	+137.25	T
Vertical	U0–L0	U8–L8	3.000	−64.05	C
	U1–L1	U7–L7	3.000	−45.75	C
	U2–L2	U6–L6	3.000	−27.45	C
	U3–L3	U5–L5	3.000	−9.15	C
	U4–L4	—	3.000	0.00	0 (zero-force)
Diagonal	U0–L1	U8–L7	4.243	+90.58	T
	U1–L2	U7–L6	4.243	+64.70	T
	U2–L3	U6–L5	4.243	+38.82	T
	U3–L4	U5–L4	4.243	+12.94	T

Figure 11: Hand calculation: joint L1 and section cut through panel 4.

Hand-solved members vs Robot (AS-P2-CAL-002 R0)

Member	Method	Hand (kN)	Robot (kN)	Difference	T / C
L0–L1	Joint L0	0.00	0.00	0.0%	Zero force
U0–L0	Joint L0	64.05	64.05	0.0%	C
U0–L1	Joint U0	90.58	90.58	0.0%	T
U0–U1	Joint U0	64.05	64.05	0.0%	C
U1–L1	Joint L1	45.75	45.75	0.0%	C
L1–L2	Joint L1	64.05	64.05	0.0%	T
U3–U4	Section cut, panel 4	146.40	146.40	0.0%	C

Reactions: 73.20 kN each by hand and in Robot (sum 146.40 kN). The other 26 members are taken from Robot and checked by symmetry and equilibrium at joint U4; they were not solved individually by hand.

Design basis

Geometry	Pratt truss, 24.0 m span, 8 panels × 3.0 m, 3.0 m deep; pin at L0, roller at L8; 33 members, statically determinate
Trial sections	Chords HSS102×102×6.4; verticals and diagonals L76×76×6.4; not design-checked
Loads	Dead 2.0 kN/m (conveyor 1.5 kN/m + grating 0.5 kPa × 1.0 m); Live 2.4 kN/m (2.4 kPa × 1.0 m); all assumed
Combination	ULS 1.25D + 1.5L = 6.1 kN/m, giving 18.3 kN at L1–L7 and 9.15 kN at L0 and L8; self-weight excluded
Scope	Member forces only. Member resistances and connections not designed.

AS-P2-CAL-002 Hand vs Robot member forces, Conveyor Gallery Truss (ULS 1.25D + 1.5L, self-weight excluded)

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Robot FX sign: + = compression; reversed below to match hand convention (+ = tension).

Bar	Member	Hand (kN)	Robot FX (kN)	Robot, hand sign (kN)	Difference (kN)	% diff	Status	Hand source
1	L0–L1	0.00	0.00	0.00	0.00	0.0%	PASS	Joints L0/U0/L1
2	L1–L2	64.05	-64.05	64.05	0.00	0.0%	PASS	Joints L0/U0/L1
3	L2–L3	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
4	L3–L4	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
5	L4–L5	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
6	L5–L6	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
7	L6–L7	64.05	-64.05	64.05	0.00	0.0%	PASS	Robot (symmetry check)
8	L7–L8	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
9	U0–U1	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
10	U1–U2	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
11	U2–U3	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
12	U3–U4	-146.40	146.40	-146.40	0.00	0.0%	PASS	Section cut, panel 4
13	U4–U5	-146.40	146.40	-146.40	0.00	0.0%	PASS	Robot (symmetry check)
14	U5–U6	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
15	U6–U7	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
16	U7–U8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
17	U0–L0	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
18	U1–L1	-45.75	45.75	-45.75	0.00	0.0%	PASS	Joints L0/U0/L1
19	U2–L2	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
20	U3–L3	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
21	U4–L4	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
22	U5–L5	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
23	U6–L6	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
24	U7–L7	-45.75	45.75	-45.75	0.00	0.0%	PASS	Robot (symmetry check)
25	U8–L8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
26	U0–L1	90.58	-90.58	90.58	0.00	0.0%	PASS	Joints L0/U0/L1
27	U1–L2	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
28	U2–L3	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
29	U3–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
30	U5–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
31	U6–L5	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
32	U7–L6	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
33	U8–L7	90.58	-90.58	90.58	0.00	0.0%	PASS	Robot (symmetry check)

Reactions: hand 73.20 kN each; Robot 73.20 kN each (sum 146.40 kN). Max force 146.40 kN = M/d check.

Members with hand source 'Robot (symmetry check)' were not individually hand-solved; they satisfy symmetry and joint U4 equilibrium.

Figure 12: AS-P2-CAL-002 hand vs Robot comparison, all 33 members.

Full documents: Appendix B, from page 28.

Project 3 – Maintenance Access in Hydropower Plants: Permanent Platforms and Scaffolding

Self-directed literature review | October 2026 | AS-P3-RPT-001 Rev 0 | Full report with references: Appendix C, page 33

6 sources reviewed (7 references cited)
3 factors compared: fall hazards, time to establish access, inspection and maintenance
5 O. Reg. 213/91 sections cited (ss. 26, 26.1, 26.3, 130, 131)
3 questions for senior engineering review

Research question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

Table 1. Information available from the reviewed sources

Reference	Relevant information	What remains unclear
Hatch [1]	Describes the McCormick platform and the problems which come with repeated scaffolding	Labour hours, incident data and maintenance requirements
Ontario guidance [2]	Explains mandatory working-at-heights training	Does not compare access systems
O. Reg. 213/91 [3]	Gives the requirements which apply to fall protection, guardrails and scaffolding	Does not say anything about replacing scaffolding with a permanent platform
UofT EHS [4]	Gives scaffold erection and inspection guidance	Does not address permanent turbine platforms
Robson et al. [5]	Reports changes in fall-related injury claims after the Ontario training requirement	Does not measure the effect of permanent platforms
ZARGES [6]	Describes quick setup of portable access equipment at a hydropower station	Not directly comparable to McCormick's permanent system

Maintenance Platform Verification

AS-P1-RPT-001

Technical Brief

Revision 0 18 October 2026

Prepared by: Abbas Saleh

Student exercise, not for construction.

Reviewed by: S. Azoubaidy, P.Eng. (comments received 2026-10-18; R1 in progress)

A 6.0 m × 3.0 m steel maintenance platform was checked using hand calculations, an Excel workbook, and a 3D Robot model. The beam and joist results agreed closely across all three methods. Differences in the post forces and several matters concerning the bracing are documented below for review.

PURPOSE AND SCOPE

I prepared this brief to record the gravity-load checks on the platform framing and compare the results from three separate methods. The work covers bending stress, shear stress, and deflection in the beams and joists, together with axial stress and elastic Euler buckling of the posts.

This is an elastic analysis exercise, not a complete structural design. CSA S16 resistance factors, lateral-torsional buckling, code column resistance, connections, and guard lateral loading have not been checked. The platform is not intended for construction.

DESIGN BASIS

Item	Basis	Item	Basis
Geometry	6.0 m × 3.0 m; top of steel 3,000 mm	Dead load	0.5 kPa grating plus self-weight
B1 main beams	W310×39; simply supported; 6.0 m span	Live load	4.8 kPa (assumed)
J1 joists	W150×22; 1.5 m spacing (5 total); 3.0 m span	Equipment load	10 kN at midspan of each B1 (assumed)
C1 posts	HSS127×127×6.4; 3.0 m; pinned; K = 1.0	Load combinations	ULS: 1.25D + 1.5L; SLS: D + L
BR1 bracing	L64×64×6.4 X-bracing on all four sides	Deflection criterion	L/360 under D + L (assumed)
Steel	CSA G40.21 350W (assumed); E = 200,000 MPa		

Section properties were taken from the CISC section database available through beamdimensions.com. The loads and criteria identified as assumed have not been supported by a cited design source.

VERIFICATION PROCEDURE

I first worked through the gravity load path by hand in Mathcad Prime (AS-P1-CAL-001), starting with the joists, then the main beams, and finally the posts. I repeated the calculations in a parametric Excel workbook (AS-P1-CAL-002). The workbook uses named input cells and includes a CHECK tab comparing the three result sets against a 2% difference criterion.

I then built a 3D frame model in Autodesk Robot Structural Analysis (AS-P1-MOD-001). Member self-weight was included, and the joist end releases were set to release both bending rotations while retaining torsional restraint. I also prepared the AutoCAD drawings (S-101 and S-201) and a Revit model (AS-P1-MOD-002, sheet S-301). The Revit schedules were used to check the modelled sections and prepare a material take-off.

ANALYSIS RESULTS

The moments, shears, and deflections for B1 and J1 were effectively identical in all three methods. The post reaction differed by 1.7%, mainly because the Robot model included brace self-weight. The post member force requires separate consideration, as explained in the findings.

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment, M_f	kN·m	13.51	13.52	13.51	0.0%	–
J1 factored shear, V_f	kN	18.02	18.02	18.02	0.0%	–
J1 service deflection	mm	3.56	3.56	3.56	0.0%	8.33 mm
B1 factored moment, M_f	kN·m	78.70	78.70	78.69	0.0%	–
B1 factored shear, V_f	kN	35.96	35.96	35.95	0.0%	–
B1 service deflection	mm	10.72	10.72	10.72	0.0%	16.67 mm
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	–
C1 Euler buckling load, P_{cr}	kN	1,546	1,546	Not run	–	–

The hand-calculated elastic stresses were 143 MPa in bending and 20 MPa in shear for B1; 85 MPa in bending and 20 MPa in shear for J1; and 16 MPa in axial stress for C1. The factored post load is approximately 3% of its Euler buckling load. These elastic stresses are below the assumed yield strength of 350 MPa, but that comparison is not a CSA S16 strength verification. The small difference between the Mathcad and Excel J1 moment values comes from using g = 9.807 m/s^2 and 9.81 m/s^2, respectively.

FINDINGS AND OUTSTANDING ITEMS

The bracing needs the most attention before any further design work. Its slenderness is high, and the current arrangement depends on how the crossing braces are connected and how they are intended to resist load. The findings below are also recorded in the REVIEW_REGISTER tab of AS-P1-MTO-001.

No.	Item	Finding and action	Status
1	C1 reaction	Robot reports 46.83 kN against 46.03 kN by hand. Its model includes approximately 2.6 kN of bracing self-weight, equivalent to roughly 0.8 kN per post at ULS. Difference explained and accepted.	Closed
2	C1 member force	The Robot post member force is 41.9 kN, versus the Robot reaction of 46.83 kN; approximately 4.9 kN reaches the support through the X-bracing in the model. The hand value is retained as the conservative check.	Open
3	BR1 slenderness	Using r_{min} = 12.5 mm calculated from I_{yp} and A, L/r is approximately 537 for the full long-brace length and 268 for the half-length. The full length exceeds both reference limits of 300 in tension and 200 in compression; the half-length satisfies the tension limit but not the compression limit. Proposed action: bolt the braces at the crossing and use a tension-only arrangement; notes added to S-101 and S-201 at R2.	Open
4	BR1 sag	Robot shows approximately 24 mm of self-weight sag in the long braces (about L/280). The proposed connection at the crossing is expected to reduce this, but it has not been verified.	Open
5	Revit model	The volume schedule exposed an incorrect W6X15 brace family, with an area around 2,950 mm^2 rather than the expected 766 mm^2. The type was corrected to the L64$\times$64$\times$6.4 equivalent and the schedule was checked again.	Closed
6	Material take-off	CISC mass-per-metre values give 1,298 kg of members. Revit volumes give 1,327 kg (+2.3%), reflecting imperial-family geometry. Applying an assumed 10% connection allowance to the CISC quantity gives 1,428 kg. The allowance remains to be confirmed.	Closed

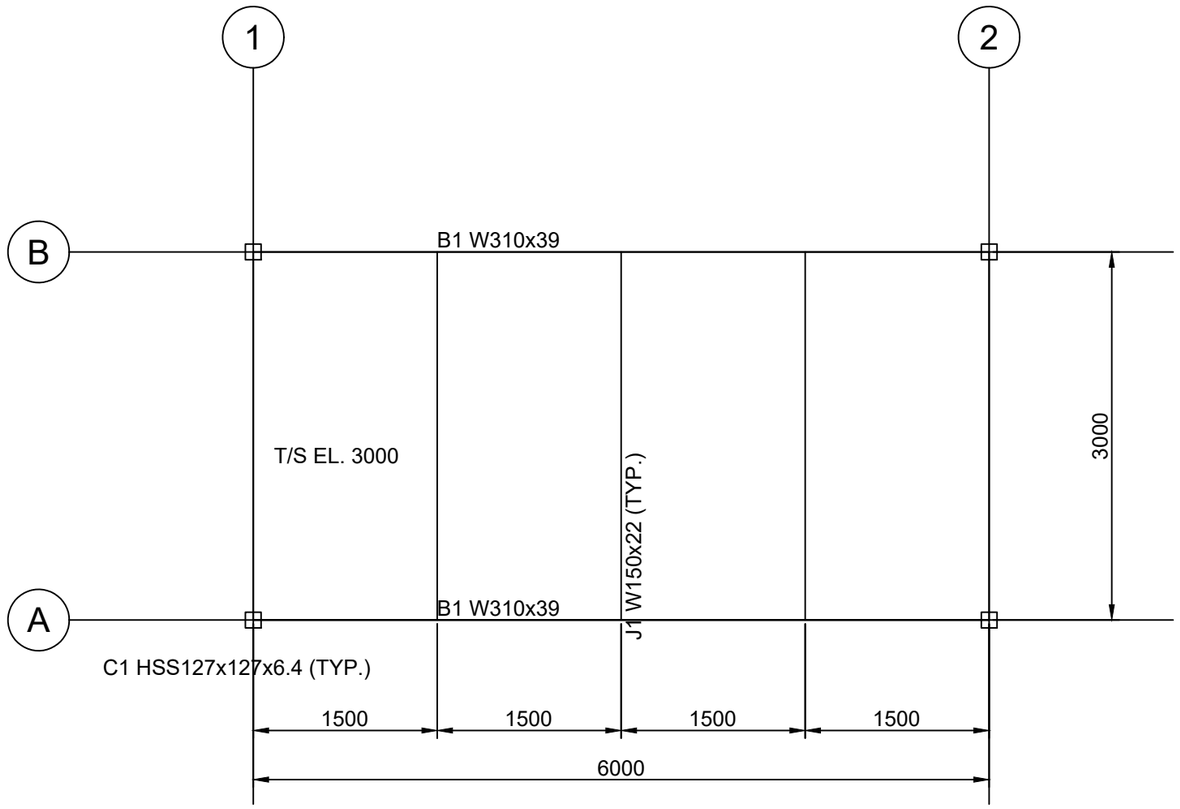
QUESTIONS FOR REVIEW

Before revising the analysis, I would appreciate feedback on the following:

1. Is a tension-only X-bracing arrangement with a bolted crossing appropriate for a platform of this size, or would a stiffer section normally be used?
2. Should the gravity load taken by the bracing be included when checking the posts, or is it normally excluded from the design model?
3. Are 4.8 kPa live load and an L/360 service deflection limit reasonable assumptions for an industrial maintenance platform? What source should be used to establish them?
4. Does the guard lateral load need to be carried through the bracing design?
5. Is a 10% allowance for connections reasonable at this stage of a material take-off?
6. What additional checks and detailing would be needed to develop this exercise into a CSA S16-compliant design?

DOCUMENTS ISSUED

Document	Description	Rev.	Document	Description	Rev.
AS-P1-CAL-001	Mathcad hand calculations	R0	S-101, S-201	AutoCAD plan, sections and notes	R2
AS-P1-CAL-002	Excel calculations and three-way comparison	R0	AS-P1-MOD-002, S-301	Revit model, schedules and sheet	R0
AS-P1-MOD-001	Robot 3D model and result plots	R0	AS-P1-MTO-001	Material take-off, slenderness check and review register	R0



X-BRACING BR1 L64x64x6.4 ALL FOUR SIDES, BOLTED AT INTERSECTION, TENSION-ONLY. SEE S-201

FRAMING PLAN 1:50

MAINTENANCE PLATFORM VERIFICATION

S-101 FRAMING PLAN

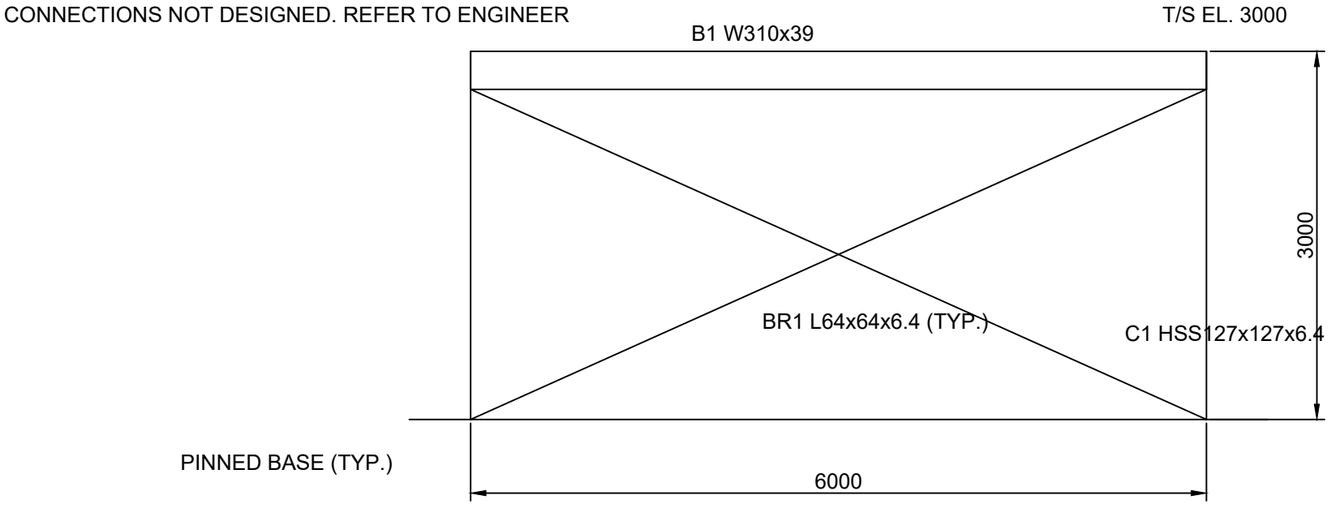
SCALE 1:50 DRAWN A. SALEH 2026-10-17

STUDENT EXERCISE. NOT FOR CONSTRUCTION

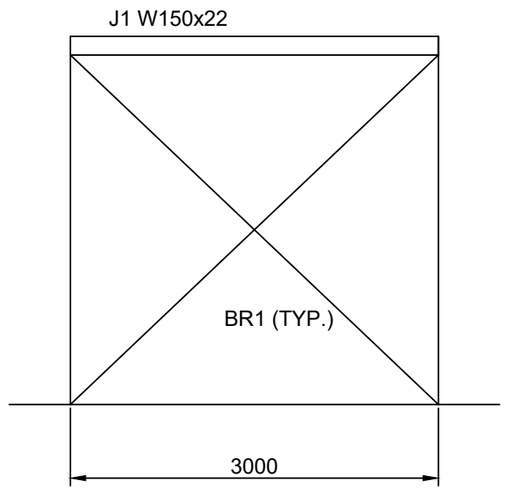
REV 0 2026-10-17 ISSUED FOR INTERNAL CHECK

REV 1 2026-10-17 ISSUED FOR REVIEW

REV 2 2026-10-18 BR1 INTERSECTION NOTE ADDED



SECTION 1 - GRID A (GRID B SIMILAR) 1:50



SECTION 2 - GRID 1 (GRID 2 SIMILAR) 1:50

GENERAL NOTES

1. STUDENT EXERCISE. NOT FOR CONSTRUCTION.
2. DESIGN BASIS: ELASTIC STRENGTH OF MATERIALS CHECK ONLY. NO CSA S16 DESIGN CHECK.
3. LOADS: GRATING 0.5 kPa + SELF-WEIGHT (DEAD); LIVE 4.8 kPa; EQUIPMENT 10 kN AT MIDSPAN OF EACH B1. ULS 1.25D + 1.5L.
4. MATERIALS: CSA G40.21 350W (ASSUMED).
5. CONNECTIONS, BRACING AND GUARD LATERAL LOAD NOT DESIGNED. REFER TO ENGINEER.
6. DIMENSIONS IN mm. ELEVATIONS IN mm ABOVE GRADE.
7. SEE CALCULATION AS-P1-CAL-001 AND ROBOT MODEL AS-P1-MOD-001.
8. BR1 BOLTED AT X-BRACE INTERSECTION. BRACING TENSION-ONLY (SLENDERNESS). SEE AS-P1-MTO-001.

MAINTENANCE PLATFORM VERIFICATION

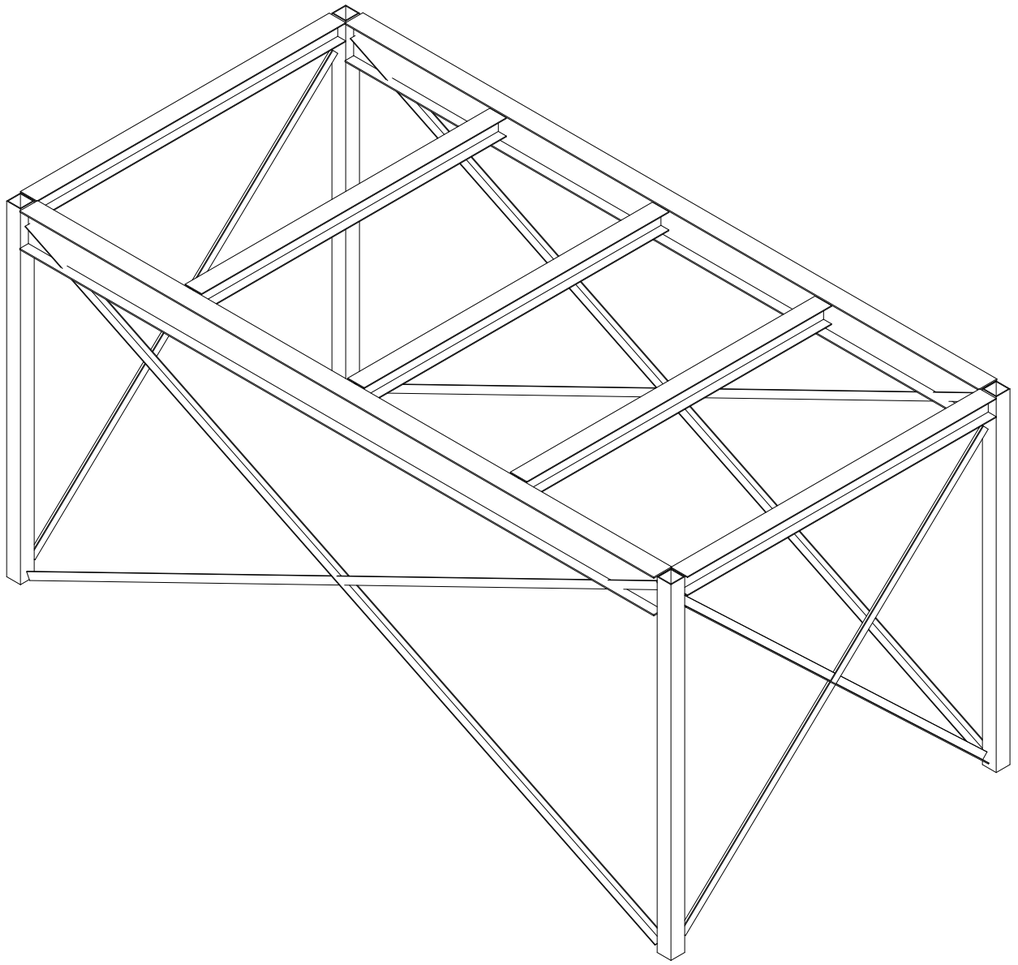
S-201 SECTIONS AND NOTES

SCALE 1:50 DRAWN A. SALEH 2026-10-17

STUDENT EXERCISE. NOT FOR CONSTRUCTION

REV 0 2026-10-17 ISSUED FOR INTERNAL CHECK
REV 1 2026-10-17 ISSUED FOR REVIEW

REV 2 2026-10-18 BR1 INTERSECTION NOTE ADDED



STUDENT EXERCISE. NOT FOR CONSTRUCTION.

P1 Steel Framing Schedule

Mark	Type	Structural Usage	Cut Length	Count	Volume
BR1-1	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-2	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-3	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-4	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-5	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-6	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-7	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-8	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
L2-1/2X2-1/2X1/4: 8			41243 mm		0.0318 m³
J1-1	W6X15	Joist	2848 mm	1	0.0084 m³
J1-2	W6X15	Joist	2810 mm	1	0.0083 m³
J1-3	W6X15	Joist	2810 mm	1	0.0083 m³
J1-4	W6X15	Joist	2810 mm	1	0.0083 m³
J1-5	W6X15	Joist	2848 mm	1	0.0084 m³
W6X15: 5			14124 mm		0.0417 m³
B1-1	W12X26	Girder	5848 mm	1	0.0301 m³
B1-2	W12X26	Girder	5848 mm	1	0.0301 m³
W12X26: 2			11695 mm		0.0601 m³
Grand total: 15			67063 mm		0.1336 m³

P1 Column Schedule

ss	Type	Base Level	Top Level	Count	System Length	Volume
C1-1	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-2	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-3	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-4	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
Grand total: 4						0.0355 m³

No.	Description	Date

Maintenance Platform Verification

Steel Model and Schedules

Project number	AS-P1
Date	2026-10-18
Drawn by	A. Saleh
Checked by	

S-301

Scale

2026-10-18 1:37:16 PM

AS-P1-CAL-001 Rev 0. Maintenance Platform Hand Calculation. Abbas Saleh, Oct 17 2026. Student exercise, not for construction.

Joist J1

$LJ := 3 \cdot m$
$sJ := 1.5 \cdot m$
$qD := 0.5 \cdot kPa$
$qL := 4.8 \cdot kPa$
$mJ := 22.5 \ \frac{kg}{m}$

$E := 200000 \cdot MPa$
$Sx := 159 \cdot 10^3 \cdot mm^3$
$Ix := 12.1 \cdot 10^6 \cdot mm^4$
$d := 152 \cdot mm$
$tw := 5.8 \cdot mm$
$wsw := mJ \cdot g$
$wD := qD \cdot sJ + wsw$
$wL := qL \cdot sJ$
$wF := 1.25 \cdot wD + 1.5 \cdot wL$

$$Mf := wF \cdot \frac{LJ^2}{8}$$
$$Vf := wF \cdot \frac{LJ}{2}$$
$$\sigma := \frac{Mf}{Sx}$$
$$\tau := \frac{Vf}{(d \cdot tw)}$$
$ws := wD + wL$

$$\delta := 5 \cdot ws \cdot \frac{LJ^4}{(384 \cdot E \cdot Ix)}$$

$$\delta lim := \frac{LJ}{360}$$

$Mf = 13.515 \ kN \cdot m$
$Vf = 18.02 \ kN$
$\sigma = 85 \ MPa$
$\tau = 20.44 \ MPa$
$\delta = 3.561 \ mm$
$\delta lim = 8.333 \ mm$

Main beam B1

$LB := 6 \cdot m$

$mB := 38.8 \cdot \frac{kg}{m}$

$SxB := 549 \cdot 10^3 \cdot mm^3$

$IxB := 85.1 \cdot 10^6 \cdot mm^4$

$dB := 310 \cdot mm$

$twB := 5.8 \cdot mm$

$Peq := 10 \cdot kN$

$wswB := mB \cdot g$

$PD := wD \cdot \frac{LJ}{2}$

$PL := wL \cdot \frac{LJ}{2}$

$Pf := 1.25 \cdot PD + 1.5 \cdot PL$

$Peqf := 1.5 \cdot Peq$

$wBf := 1.25 \cdot wswB$

$RB := \frac{(3 \cdot Pf + Peqf)}{2} + wBf \cdot \frac{LB}{2}$

$MB := RB \cdot 0.5 \cdot LB - Pf \cdot 1.5 \cdot m - 0.125 \cdot wBf \cdot LB^2$

$\sigma B := \frac{MB}{SxB}$

$\tau B := \frac{RB}{(dB \cdot twB)}$

$Ps := PD + PL$

$a1 := 1.5 \cdot m$

$\delta 1 := 2 \cdot Ps \cdot a1 \cdot \frac{\langle 3 \cdot LB^2 - 4 \cdot a1^2 \rangle}{(48 \cdot E \cdot IxB)}$

$\delta 2 := (Ps + Peq) \cdot \frac{LB^3}{(48 \cdot E \cdot IxB)}$

$\delta 3 := 5 \cdot wswB \cdot \frac{LB^4}{(384 \cdot E \cdot IxB)}$

$\delta B := \delta 1 + \delta 2 + \delta 3$

$\delta Blim := \frac{LB}{360}$

$RB = 35.957\ \boldsymbol{kN}$
$MB = 78.7\ \boldsymbol{kN \cdot m}$
$\sigma B = 143.352\ \boldsymbol{MPa}$
$\tau B = 19.998\ \boldsymbol{MPa}$
$\delta B = 10.717\ \boldsymbol{mm}$
$\delta Blim = 16.667\ \boldsymbol{mm}$

Post C1

$HC := 3 \cdot \boldsymbol{m}$
$K := 1$
$AC := 2960 \cdot \boldsymbol{mm}^2$
$IC := 7.05 \cdot 10^6 \cdot \boldsymbol{mm}^4$

$mC := 23.2 \cdot \dfrac{\boldsymbol{kg}}{\boldsymbol{m}}$

$wDe := qD \cdot 0.75 \cdot \boldsymbol{m} + wsw$
$wLe := qL \cdot 0.75 \cdot \boldsymbol{m}$

$Re := \left(1.25 \cdot wDe + 1.5\ wLe\right) \cdot 0.5 \cdot LJ$
$PC := RB + Re + 1.25 \cdot mC \cdot g \cdot HC$

$Pcr := \pi^2 \cdot E \cdot \dfrac{IC}{\left(K \cdot HC\right)^2}$

$\sigma C := \dfrac{PC}{AC}$

$ratioC := \dfrac{PC}{Pcr}$

$PC = 46.027\ \boldsymbol{kN}$
$Pcr = \left(1.546 \cdot 10^3\right)\ \boldsymbol{kN}$
$\sigma C = 15.55\ \boldsymbol{MPa}$
$ratioC = 0.03$

INPUT AS-P1-INP-001 Input Sheet

GEOMETRY	Symbol	Value	Unit	Source / assumption
Main beam span (grid 1 to 2)	L_B1	6	m	Assumed scenario
Joist span (grid A to B)	L_J1	3	m	Assumed scenario
Joist spacing	s_J	1.5	m	Assumed scenario
Number of joists	n_J	5	-	At 0, 1.5, 3.0, 4.5, 6.0 m
Post height (grade to top of steel)	H_C1	3	m	Assumed scenario
Post effective length factor	K	1	-	Assumed: braced frame, pinned-pinned posts
Equipment load position on B1	x_eq	3	m	Midspan of B1

MATERIAL	Symbol	Value	Unit	Source / assumption
Yield strength	Fy	350	MPa	Assumed CSA G40.21 350W
Elastic modulus	E	200000	MPa	Standard value for structural steel
Gravity	g	9.81	m/s2	Standard

SECTION PROPERTIES (fill from CISC database)	Symbol	Value	Unit	Source / assumption
B1 W310x39: area	A	4940	mm2	beamdimensions.com, CISC W310x39
B1 W310x39: depth	d	310	mm	beamdimensions.com, CISC W310x39
B1 W310x39: web thickness	w	5.8	mm	beamdimensions.com, CISC W310x39
B1 W310x39: moment of inertia (strong axis)	Ix	85.1	x10^6 mm4	beamdimensions.com, CISC W310x39
B1 W310x39: elastic section modulus	Sx	549	x10^3 mm3	beamdimensions.com, CISC W310x39
B1 W310x39: mass	m	38.8	kg/m	beamdimensions.com, CISC W310x39
B1 self-weight (formula)	w_sw	0.380628	kN/m	Formula: mass x g / 1000
J1 W150x22: area	A	2860	mm2	beamdimensions.com, CISC W150x22
J1 W150x22: depth	d	152	mm	beamdimensions.com, CISC W150x22
J1 W150x22: web thickness	w	5.8	mm	beamdimensions.com, CISC W150x22
J1 W150x22: moment of inertia (strong axis)	Ix	12.1	x10^6 mm4	beamdimensions.com, CISC W150x22

J1 W150x22: elastic section modulus	Sx	159	x10^3 mm3	beamdimensions.com, CISC W150x22
J1 W150x22: mass	m	22.5	kg/m	beamdimensions.com, CISC W150x22
J1 self-weight (formula)	w_sw	0.220725	kN/m	Formula: mass x g / 1000
C1 HSS127x127x6.4: area	A	2960	mm2	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: moment of inertia	I	7.05	x10^6 mm4	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: radius of gyration	r	48.8	mm	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: mass	m	23.2	kg/m	CISC HSS table (search HSS127x127x6.4 on the same site)
BR1 L64x64x6.4: area	A	766	mm2	CISC angle table (bracing: model + MTO only, not designed)
BR1 L64x64x6.4: mass	m	6.01	kg/m	CISC angle table (bracing: model + MTO only, not designed)

LOADS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Grating (dead)	q_D	0.5	kPa	Assumed
Floor live load	q_L	4.8	kPa	Assumed; replace with a cited value if found
Equipment point load (live), each main beam	P_eq	10	kN	Assumed
Guard load	-	cited only	-	Cite NBCC 2020 Div. B Part 4 guard clause / O. Reg. 213/91 guardrail section. NOT designed; flagged for review. Required per review R4; to be added at R1.
Interior joist tributary width	b_int	1.5	m	Formula: joist spacing
End joist tributary width	b_end	0.75	m	Formula: half joist spacing

LOAD COMBINATIONS / LIMITS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Dead load factor (strength)	aD	1.25	-	1.25D + 1.5L
Live load factor (strength)	aL	1.5	-	1.25D + 1.5L
Deflection limit denominator (service, D+L)	L/	360	-	L/360 assumed

SCOPE LIMITS

1. Elastic Strength of Materials analysis only.
2. No CSA S16 resistance factors, no lateral-torsional buckling check, no column curve.
3. Connections, bracing and guard lateral load are not designed; flagged for senior review.
4. Loads are assumed unless a source is cited in column E.
5. Student exercise. Not for construction.

CHECK Three-way comparison

Quantity	Unit	Mathcad (hand)	Excel	Robot	% diff Excel vs hand	% diff Robot vs hand	PASS ($\leq$2%)	Cause if $>$2%
J1 max factored moment Mf	kN·m	13.51	13.515	13.51	0.0%	0.0%	PASS	Mathcad g = 9.807 m/s^2, Excel g = 9.81 m/s^2; under 0.05% effect on self-weight terms.
J1 max factored shear Vf	kN	18.02	18.020	18.02	0.0%	0.0%	PASS	
J1 service deflection	mm	3.561	3.561	3.56	0.0%	0.0%	PASS	
B1 reaction (factored)	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 max factored moment Mf	kN·m	78.7	78.701	78.69	0.0%	0.0%	PASS	
B1 max factored shear Vf	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 service deflection	mm	10.72	10.717	10.72	0.0%	0.0%	PASS	
C1 axial load (factored)	kN	46.03	46.028	46.83	0.0%	1.7%	PASS	Robot includes bracing self-weight (2.6 kN total; DL1 total 22.17 kN vs. 19.6 kN hand); +0.8 kN per post at ULS.
C1 Euler buckling load Pcr	kN	1546	1546.238	not run in robot	0.0%	N/A	N/A	
Total dead load	kN	19.6		22.17				bracing
Total live load	kN	106.4		106.4				

Review flags

- Long-side L64 braces sag about 24 mm under self-weight (L/280). Flag for senior review: consider a larger angle or a mid-length tie
- C1 member axial 41.9 kN (Robot) vs. 46.0 kN (hand): X-bracing shares about 4.9 kN of gravity load; the hand calc is conservative.

CALC Excel calculation (named inputs only)

J1 JOIST (interior)

Quantity	Value	Unit
wD dead UDL	0.971	kN/m
wL live UDL	7.200	kN/m
wf factored UDL	12.013	kN/m
Mf	13.515	kN·m
Vf	18.020	kN
sigma	85.001	MPa
tau	20.440	MPa
deflection (service)	3.561	mm
limit L/360	8.333	mm

B1 MAIN BEAM

Quantity	Value	Unit
PD per joist end	1.456	kN
PL per joist end	10.800	kN
Pf factored joist load	18.020	kN
Peq factored	15.000	kN
self-weight factored	0.476	kN/m
RB reaction	35.958	kN
MB midspan moment	78.701	kN·m
sigma	143.354	MPa
tau	19.999	MPa
Ps service joist load	12.256	kN
defl: joists at 1.5 and 4.5 m	4.456	mm
defl: midspan joist + equipment	5.884	mm
defl: self-weight	0.377	mm
deflection total	10.717	mm
limit L/360	16.667	mm

C1 POST (corner)

Quantity	Value	Unit
wD end joist	0.596	kN/m
wL end joist	3.600	kN/m
Re end joist reaction	9.217	kN
PC axial (factored)	46.028	kN
Pcr Euler	1546.238	kN
sigma axial	15.550	MPa
PC / Pcr	0.030	-

Values shown to 3 decimals; the live workbook (AS-P1-CAL-002.xlsx) holds full precision.

AS-P1-MTO-001 Material Take-Off: Maintenance Platform Verification

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Lengths from Revit schedules (AS-P1-MOD-002, sheet S-301); mass/m from CISC section tables.

Mark	Member	Section (CISC)	Revit type	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
B1	Main beam	W310x39	W12X26	2	5.848	11.696	38.8	453.8	Revit cut length
J1 (end)	Joist	W150x22	W6X15	2	2.848	5.696	22.5	128.2	J1-1, J1-5
J1 (int.)	Joist	W150x22	W6X15	3	2.810	8.430	22.5	189.7	J1-2 to J1-4
C1	Post	HSS127x127x6.4	HSS5X5X1/4	4	3.000	12.000	23.2	278.4	System length GRADE to T/S
BR1 (long)	X-brace, long sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	6.463	25.852	6.01	155.4	BR1-1 to BR1-4
BR1 (short)	X-brace, short sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	3.848	15.392	6.01	92.5	BR1-5 to BR1-8
Subtotal, members				**19**		**79.066**		**1297.9**	
Connection allowance	Plates, bolts, welds						10%	129.8	Preliminary estimate, accepted by reviewer (R5). Replace with detailed connection weights later.
TOTAL STEEL								**1427.7**	**kg**

Cross-check against Revit schedule volumes

Item	Value	Unit	Source
Framing volume	0.1336	m3	P1 Steel Framing Schedule, grand total
Column volume	0.0355	m3	P1 Column Schedule, grand total
Steel density	7850.0	kg/m3	Standard value for structural steel
Mass from Revit volume	1327.4	kg	Formula
Mass from CISC kg/m	1297.9	kg	Members subtotal above
Difference	2.3%		Expected small: Revit families use imperial section geometry
Status	WITHIN 5% - OK		Tolerance 5% (own criterion)

AS-P1-MTO-001 BR1 slenderness check, L64x64x6.4 X-bracing

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Lengths are work-point geometry, conservative vs Revit cut length.

r_min (mm)	12.5	Computed: √(Iyp/A), Iyp and A from beamdimensions.com L64x64x6.4
Tension limit L/r	300	CSA S16 slenderness limit, tension members
Compression limit KL/r	200	CSA S16 slenderness limit, compression members

Brace	Panel w (mm)	Panel h (mm)	L full (mm)	L/r full	L at crossing (mm)	L/r at crossing	Tension-only check	If designed for compression
Long sides (BR1-1 to 4)	6000	3000	6708	537	3354	268	OK if bolted at crossing	FAIL - keep tension-only
Short sides (BR1-5 to 8)	3000	3000	4243	339	2121	170	OK if bolted at crossing	OK

Finding: Full-length L/r exceeds the tension limit. Intersection note added to S-101/S-201 at R2. Per review R1, a crossing connection does not by itself confirm restraint; to be verified with a tension-only lateral analysis.

AS-P1-MTO-001 Review and Issue Register

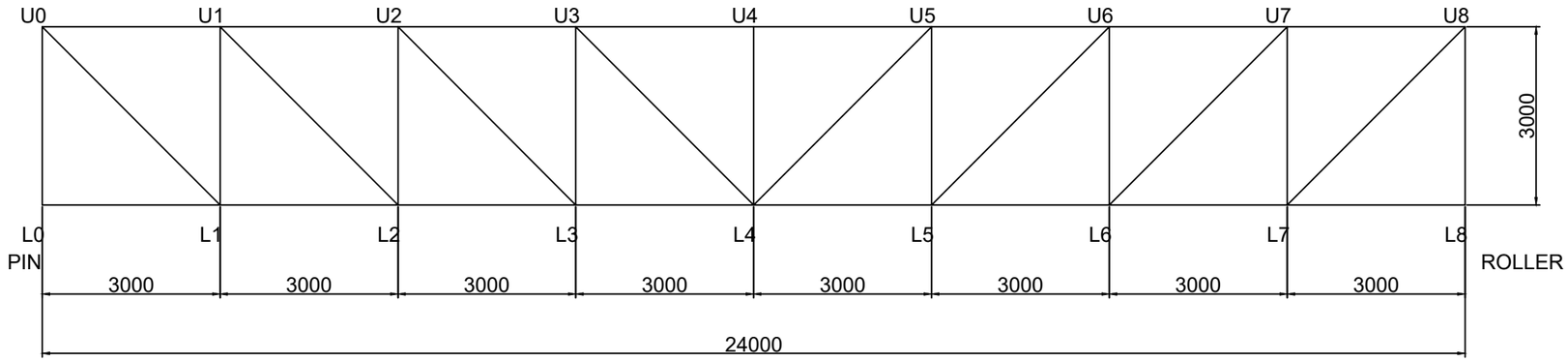
Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction.

No.	Date	Source	Item	Finding	Action	Status
1	2026-10-18	Revit schedule	BR1 section	Braces modelled as W6X15. Schedule volume check gave ~2950 mm2 area vs 766 mm2 expected for L64x64x6.4.	Type changed to L2-1/2X2-1/2X1/4 (L64x64x6.4 equivalent). Schedule re-checked.	Closed
2	2026-10-18	Revit schedule	J1 structural usage	Joists tagged as Girder.	Changed to Joist.	Closed
3	2026-10-18	Revit model	BR1 placement method	Braces drawn with Beam tool, so Structural Usage cannot be set to Vertical Bracing.	Accepted for this model. Use Brace tool on future models.	Closed
4	2026-10-17	Drawing S-101 R0	C1 label	Section labelled HSS128x128.	Corrected to HSS127x127x6.4 in R1.	Closed
5	2026-10-17	Drawing S-101 R0	BR1 label	Typo 'BR!'.	Corrected to BR1 in R1.	Closed
6	2026-10-17	Drawing S-101 R0	T/S label	Top-of-steel label poorly placed.	Moved in R1.	Closed
7	2026-10-18	Robot vs hand calc	C1 reaction	Robot 46.83 kN vs hand 46.03 kN.	Difference = bracing self-weight, included in Robot only. Accepted.	Closed
8	2026-10-18	Robot vs hand calc	C1 axial force	Robot member force 41.9 kN, lower than hand calc.	Bracing shares gravity load in the 3D model. Hand calc is conservative. Note in brief.	Open
9	2026-10-18	Slenderness check	BR1 L/r	Full-length L/r exceeds tension limit (see SLENDERNESS tab).	Bolt braces at X intersection, tension-only design. Add note to S-101 and brief.	Open
10	2026-10-18	Robot	BR1 self-weight sag	Brace sag ~24 mm (about L/280) under self-weight.	Intersection connection expected to reduce it; not verified. Mention in brief.	Open
11	2026-10-18	MTO	Mass cross-check	Revit volume mass vs CISC kg/m mass differ by a few %.	Imperial family geometry. Within 5% tolerance. Accepted.	Closed
12	2026-10-18	Revit type	C1 section area	Earlier note: Revit HSS area 2774 mm2 vs CISC 2960 mm2.	Checked: column schedule volume 0.0089 m3 / 3.0 m = 2967 mm2, matches CISC 2960 mm2. No	Closed

External review comments

No.	Date	Reviewer	Comment (verbatim)	Response	Action	Status
R1	2026-10-18	S. Azoubaidy, P.Eng.	"I'd suggest modelling the braces as tension-only members, applying the relevant lateral loads, and checking the resulting member forces and displacements."	Accepted. To be addressed at R1.	Set BR1 tension-only in Robot; add lateral load case	Open
R2	2026-10-18	S. Azoubaidy, P.Eng.	"What I would check first is whether Robot is allowing any of the braces to carry compression when you've intended them to act in tension only."	Accepted. To be addressed at R1.	Re-run with tension-only braces; explain the 41.9 vs 46.03 kN gap. Keep 46.03 kN for now.	Open
R3	2026-10-18	S. Azoubaidy, P.Eng.	"For your next revision, I'd suggest identifying the relevant code provisions and explaining why you've selected those criteria."	Accepted. To be addressed at R1.	Cite the OBC live-load and deflection provisions in the design basis	Open
R4	2026-10-18	S. Azoubaidy, P.Eng.	"I'd include the guard loading in your lateral analysis and then check the resulting member forces and connections."	Accepted. To be addressed at R1.	Add guard load case (OBC); check brace forces and drift	Open

R5	2026-10-18	S. Azoubaidy, P.Eng.	"I don't have an issue with using 10% as a preliminary allowance for connections at this stage."	Accepted. Allowance kept and labelled as an estimate.	Keep it labelled as an estimate	Closed
R6	2026-10-18	S. Azoubaidy, P.Eng.	"Just keep in mind that they don't replace the resistance checks required by CSA S16."	Accepted. To be addressed at R1.	Add his S16 checklist to the brief as "Not yet checked"	Open



CONVEYOR GALLERY TRUSS - ELEVATION 1:100

1. TRIAL SECTIONS: CHORDS HSS102x102x6.4; VERTICALS AND DIAGONALS L76x76x6.4. NOT DESIGN-CHECKED.
2. LOADS (ASSUMED): DEAD 2.0 kN/m, LIVE 2.4 kN/m. ULS 1.25D + 1.5L = 18.3 kN AT L1-L7.
3. MEMBER FORCES: SEE AS-P2-CAL-001 (HAND CHECK) AND AS-P2-CAL-002 (VERIFIED AGAINST ROBOT AS-P2-MOD-001).
4. CONNECTIONS NOT DESIGNED. STUDENT EXERCISE. NOT FOR CONSTRUCTION.

CONVEYOR GALLERY TRUSS VERIFICATION

S-301 TRUSS ELEVATION

SCALE 1:100 DRAWN A. SALEH 2026-10-18

STUDENT EXERCISE. NOT FOR CONSTRUCTION

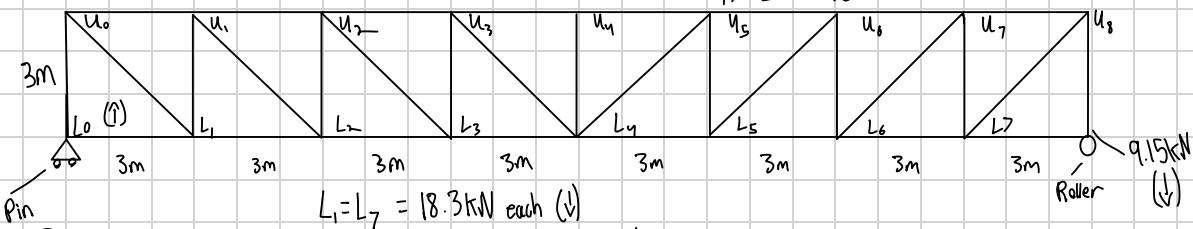
REV 0 2026-10-18 FIRST ISSUE

P2 Conveyor Gallery Truss

Sign Convention: Tension (+), Compression (-)

73.2 kN (↑) at L0 and L8

No loads U0-U8. Horizontal reaction L0 = 0



Determinacy check

33 members + 3 reactions = 36 = 2x18 nodes.

Therefore the truss is determinate.

Assumed Loads

Dead: 1.5 kN/m conveyor + 0.5 kPa x 1m grating = 2 kN/m

Live: 2.4 kPa x 1.0 m = 2.4 kN/m

ULS: 1.25x2 + 1.5x2.4 = 6.1 kN/m

Nodal loads: 18.3kN down at L1-L7, and 9.15 kN at L0 and L8

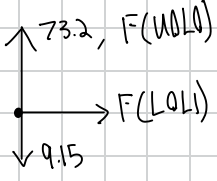
Diagonal Length 4.243m, Sin = Cos = 0.7071

Reactions

Total load = 6.1 x 24 = 146.4 kN

Each reaction = $\frac{146.4}{2}$ = 73.2

Joint L0



$\sum F_x = 0$: F(L0L1) = 0, so L0-L1 = 0 (zero force)

$\sum F_y = 0$: 73.2 - 9.15 + F(U0L0) = 0, so F(U0L0) = -64.05, which is 64.05 kN (C)

Joint U0



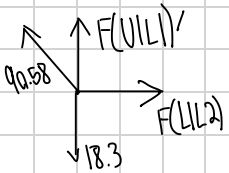
$\sum F$

$\sum F_y = 0$: 64.05 - 0.7071 · F(U0L1) = 0, So F(U0L1) = 90.58 kN (T)

$\sum F_x = 0$: F(U0U1) + 0.707 (90.58) = 0, So F(U0U1) = -64.05 kN

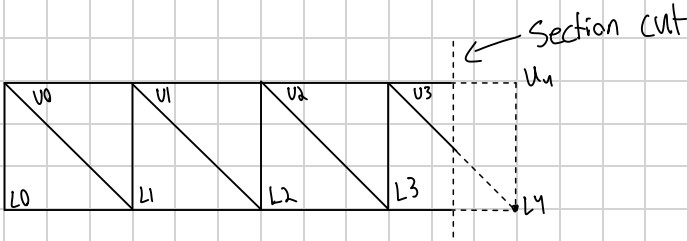
= 64.05 kN (C)

Joint L1



$\sum F_y = 0: 0.7071(90.58) + F(U1L1) - 18.3 = 0$, So $F(U1L1) = -45.75 \rightarrow 45.75\,kN(C)$

$\sum F_x = 0: -0.7071(90.58) + F(L1L2) = 0$, So $F(L1L2) = 64.05\,kN(T)$



$\sum M_{L4} = 0: 73.2(12) - 9.15(12) - 18.3(9+6+3) - F(U3U4)\cdot(-3) = 0$

$F(U3U4) = \frac{-439.2}{3} = -146.40 \rightarrow 146.40\,kN(C)$

Full member force table (hand results above verified against robot model)

Group	Member	Mirror (right half)	Length (m)	Force (kN)	T / C
Top chord	U0–U1	U7–U8	3.000	−64.05	C
	U1–U2	U6–U7	3.000	−109.80	C
	U2–U3	U5–U6	3.000	−137.25	C
	U3–U4	U4–U5	3.000	−146.40	C
Bottom chord	L0–L1	L7–L8	3.000	0.00	0 (zero-force)
	L1–L2	L6–L7	3.000	+64.05	T
	L2–L3	L5–L6	3.000	+109.80	T
	L3–L4	L4–L5	3.000	+137.25	T
Vertical	U0–L0	U8–L8	3.000	−64.05	C
	U1–L1	U7–L7	3.000	−45.75	C
	U2–L2	U6–L6	3.000	−27.45	C
	U3–L3	U5–L5	3.000	−9.15	C
	U4–L4	—	3.000	0.00	0 (zero-force)
Diagonal	U0–L1	U8–L7	4.243	+90.58	T
	U1–L2	U7–L6	4.243	+64.70	T
	U2–L3	U6–L5	4.243	+38.82	T
	U3–L4	U5–L4	4.243	+12.94	T

AS-P2-CAL-002 Hand vs Robot member forces, Conveyor Gallery Truss (ULS 1.25D + 1.5L, self-weight excluded)

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Robot FX sign: + = compression; reversed below to match hand convention (+ = tension).

Bar	Member	Hand (kN)	Robot FX (kN)	Robot, hand sign (kN)	Difference (kN)	% diff	Status	Hand source
1	L0–L1	0.00	0.00	0.00	0.00	0.0%	PASS	Joints L0/U0/L1
2	L1–L2	64.05	-64.05	64.05	0.00	0.0%	PASS	Joints L0/U0/L1
3	L2–L3	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
4	L3–L4	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
5	L4–L5	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
6	L5–L6	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
7	L6–L7	64.05	-64.05	64.05	0.00	0.0%	PASS	Robot (symmetry check)
8	L7–L8	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
9	U0–U1	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
10	U1–U2	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
11	U2–U3	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
12	U3–U4	-146.40	146.40	-146.40	0.00	0.0%	PASS	Section cut, panel 4
13	U4–U5	-146.40	146.40	-146.40	0.00	0.0%	PASS	Robot (symmetry check)
14	U5–U6	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
15	U6–U7	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
16	U7–U8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
17	U0–L0	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
18	U1–L1	-45.75	45.75	-45.75	0.00	0.0%	PASS	Joints L0/U0/L1
19	U2–L2	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
20	U3–L3	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
21	U4–L4	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
22	U5–L5	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
23	U6–L6	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
24	U7–L7	-45.75	45.75	-45.75	0.00	0.0%	PASS	Robot (symmetry check)
25	U8–L8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
26	U0–L1	90.58	-90.58	90.58	0.00	0.0%	PASS	Joints L0/U0/L1
27	U1–L2	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
28	U2–L3	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
29	U3–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
30	U5–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
31	U6–L5	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
32	U7–L6	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
33	U8–L7	90.58	-90.58	90.58	0.00	0.0%	PASS	Robot (symmetry check)

Reactions: hand 73.20 kN each; Robot 73.20 kN each (sum 146.40 kN). Max force 146.40 kN = M/d check.

Members with hand source 'Robot (symmetry check)' were not individually hand-solved; they satisfy symmetry and joint U4 equilibrium.

AS-P2-MTO-001 Material Take-Off: Conveyor Gallery Truss Verification

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Based on S-301 Rev 0; trial sections; centreline lengths. Student exercise, not for construction.

Mark	Member	Section (CISC)	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
TC	Top chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	U0-U1 to U7-U8
BC	Bottom chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	L0-L1 to L7-L8
V	Vertical	L76x76x6.4	9	3.000	27.00	7.28	196.6	U0-L0 to U8-L8
D	Diagonal	L76x76x6.4	8	4.243	33.94	7.28	247.1	U0-L1 to U3-L4, U5-L4 to U8-L7
Subtotal, one truss			**33**		**108.94**		**1317.3**	
Connection allowance	Gusset plates, bolts, welds					10%	131.7	10% as in AS-P1-MTO-001; replace once connections designed
TOTAL, ONE TRUSS							**1449.0**	**kg**
TOTAL, TWO TRUSSES			**2**				**2898.0**	**kg (assumed one truss each side of gallery)**

Checks

Item	Value	Target	Status
Member count	33	33	OK - matches 2j - r = 33
Chord length, each chord (m)	24.00	24	OK - equals span
Vertical count	9	9	OK - 8 panels + 1
Diagonal length (m)	4.243	4.243	OK - 3 m panel x 3 m depth

Mass per metre = A x 7850 kg/m3, with A from CISC section properties (HSS102x102x6.4: 2320 mm2; L76x76x6.4: 927 mm2), same method as AS-P1-MTO-001. Lengths are node-to-node centreline lengths from S-301; no allowance for cut lengths or end preparation.

Maintenance Access in Hydropower Plants: Permanent Platforms and Scaffolding

AS-P3-RPT-001

Revision 0 18 October 2026

Literature Review Prepared by: Abbas Saleh

Research Question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

1. Background

In many cases, inspecting the underside of a turbine runner is more of an access problem than an inspection problem. The work has to be done inside the draft tube, where there is not much room to move around. The usual approach is to build scaffolding inside the draft tube elbow so that the inspection areas can then be reached. After the work is finished, the scaffold is then removed. If access is needed again during a later outage, the same arrangement then has to be built again from the beginning.

Hatch (2026) published a different approach in January 2026 [1]. A modular access platform was developed for SCHM's McCormick Hydropower Plant in Québec. The project started in 2019 after a safety concern was found underneath the turbine runner during an inspection. From the information which Hatch provided, the finished system serves five turbine-generator units, works with three different turbine types, and removes the need for conventional scaffolding.

The reason this project was chosen is because turbine maintenance is repeated. A temporary solution may be fine when access is only needed once in a while. It becomes a lot less attractive when essentially the same working area has to be set up again every time the equipment is inspected. At the same time, this does not automatically mean that the permanent installation is the better option. It still has to be designed for the site, inspected, and also maintained. Because of this, what needs to be looked at is whether the amount of scaffold work which is avoided is actually worth more than the extra responsibilities which then come with the permanent platform.

2. Sources and Scope

Six publicly available sources were reviewed. These include Ontario safety legislation, provincial working-at-heights guidance, a University of Toronto scaffold standard, one peer-reviewed study, and project information which was published by Hatch and by ZARGES. The legislation and safety documents were used to find the requirements which apply to scaffolding and fall protection. The Hatch and ZARGES publications were then

used to look at the alternatives to conventional access which are currently being used in hydropower stations.

There is a limitation with using company publications for this kind of comparison. The companies explain what their own systems are supposed to do. At the same time, the operating records which would be needed to actually check those benefits are not included, so the claims cannot be confirmed from the publications on their own. Ontario Regulation 213/91 was also used as the reference regulation. This is because it has specific requirements in it for scaffolding and also for fall protection. It is not being presented as the regulation which governs McCormick, since McCormick is located in Québec. This review also did not try to verify the structural capacity of either access system or estimate the cost of building a permanent platform. Both of these would have to be done separately.

Table 1. Information available from the reviewed sources

Reference	Relevant information	What remains unclear
Hatch [1]	Describes the McCormick platform and the problems which come with repeated scaffolding	Labour hours, incident data and maintenance requirements
Ontario guidance [2]	Explains mandatory working-at-heights training	Does not compare access systems
O. Reg. 213/91 [3]	Gives the requirements which apply to fall protection, guardrails and scaffolding	Does not say anything about replacing scaffolding with a permanent platform
UofT EHS [4]	Gives scaffold erection and inspection guidance	Does not address permanent turbine platforms
Robson et al. [5]	Reports changes in fall-related injury claims after the Ontario training requirement	Does not measure the effect of permanent platforms
ZARGES [6]	Describes quick setup of portable access equipment at a hydropower station	Not directly comparable to McCormick's permanent system

3. Discussion

3.1 Fall Hazards

When scaffolding is being considered, it is easy to only think about the finished platform which the workers end up standing on. However, someone then has to build it first. During erection, workers may be at height before the whole scaffold and its protective systems are complete. The same problem then comes up again during dismantling. These activities are separate from the turbine inspection, but they still expose workers to hazards just so that the inspection can take place.

There are a few parts of Ontario Regulation 213/91 which then apply to this kind of work [3]. In Section 26, fall protection is required where a worker could fall more than three metres. In Section 26.3, guardrails are required along scaffold and platform edges where the distance which the worker could fall is 2.4 metres or more. Section 26.1 then also states that guardrail systems should be used ahead of the personal fall protection methods. This difference matters here, since a worker who is standing on a finished and properly guarded platform is in a very different situation from a worker who is assembling access equipment while some of that protection is not in place yet. The University of Toronto scaffold standard also covers protection for workers who are doing erection work at height for the same reason [4]. In addition, the mandatory working-at-heights training in Ontario applies to workers who use fall protection devices such as travel restraint and fall arrest systems, and guardrails are not included in that list [2], which is consistent with the preference for guardrail systems which is given in Section 26.1 [3].

Taking this into consideration, a permanent platform with proper guarding could avoid the repeated exposure which happens during scaffold assembly and removal. This would be a meaningful advantage for a station

which needs access during several outages. It would not make the work free of fall hazards though. The platform would still need to provide safe access, proper edge protection, and also protection for any activity which takes a worker outside of the guarded area.

Robson et al. (2024) give some useful background on fall prevention [5]. Their study examined the mandatory working-at-heights training in Ontario. From their results, the lost-time claim rate for targeted fall injuries was 19% lower in 2017–19 than in 2012–14, compared to a 6% reduction in other provinces over the same period. These percentages should not be taken as the expected improvement from installing a permanent platform, since the study looked at training and not at alternative access structures. The more defensible conclusion is a narrower one. Avoiding scaffold erection and dismantling removes two recurring activities where fall exposure can happen. The actual reduction in risk would then have to be worked out for the specific station and work procedure.

3.2 Time Required to Establish Access

Time becomes a more important consideration when the same access equipment has to be installed again and again. For a scaffold arrangement, the work does not start when the inspection crew steps onto the platform. The components first have to be brought into position, assembled, and then inspected. Once the maintenance is finished, the structure then has to be dismantled and removed as well. At a generating station this work also has to fit within the planned outage. For this reason, the time which is spent setting up access is worth looking at separately from the inspection itself.

Hatch (2026) identifies scaffold installation as a time-consuming part of the conventional method [1]. However, the article does not give the crew size, the number of hours which are normally needed, or the effect on the outage schedule. Because of this, the size of the saving at McCormick cannot be worked out from it.

ZARGES gives another example from the Töging am Inn hydropower station in Germany [6]. The station has fourteen turbines, and one unit goes through a major overhaul each year over approximately three months. Portable work platforms were used in place of scaffolding for components up to four metres high, and ZARGES states that the equipment can be ready to use in less than a minute. This number needs some context. The ZARGES equipment is a portable access system. It is not the same as the permanent modular arrangement which was developed for McCormick, and so the setup time cannot be carried over from one system to the other. Both projects show why alternatives to scaffolding are being considered, but neither one gives enough data to calculate the time which would actually be saved.

A useful comparison would start from the existing maintenance procedure. Some examples of what would be needed are the number of workers who are assigned to scaffold erection, how long the installation takes, how often it is repeated, and whether the scaffold removal affects when the generating unit can go back into service. Once these figures are available, the setup and removal time for the permanent system can then be compared against the existing approach. Without them, a numerical claim about improved efficiency would be difficult to support.

3.3 Inspection and Maintenance

The inspection requirements are where the comparison becomes less straightforward. Temporary scaffolding has to be checked before workers use it. Ontario Regulation 213/91 covers scaffold design and inspection in Section 130 and the supervision of erection, alteration, and dismantling in Section 131 [3]. Under Section 130(1), a scaffold exceeding 15 m above its base, or 10 m for a tube-and-clamp scaffold, is also subject to additional design requirements. The University of Toronto standard also requires scaffold inspections after erection and after relocation [4]. Since the scaffold is assembled again and again, many of these activities then have to be repeated during each maintenance outage.

A permanent platform would not have the same recurring erection process. However, it would bring long-term inspection responsibilities of its own. This is particularly relevant inside a draft tube. The moisture and the operating environment mean that corrosion protection has to be considered, and the connections, supporting members, and also the guardrails would all have to stay in acceptable condition over the service life of the platform. The available publications do not explain how the McCormick platform is maintained. The

inspection interval, intended design life, protective coating system, and whether inspection responsibility sits with the station owner or with another party could not be found. Because of this, a fair comparison of the total maintenance effort is not possible from the information which is available. It would also be premature to conclude that the permanent platform needs less inspection just because it does not have to be erected for every outage. Its inspections may be less frequent, but their scope and complexity still have to be determined.

4. Limitations of the Review

Several points would need to be resolved before this research could be used to support an actual project decision.

The first is the applicable legislation. McCormick is located in Québec, and the relevant Québec requirements were not reviewed. Ontario Regulation 213/91 is a useful reference for discussing scaffolding, but it cannot be assumed to apply to the McCormick installation. There is also a further distinction for an Ontario station. Depending on the type of work, the applicable requirements may fall under O. Reg. 213/91 for construction projects [3] or under Reg. 851 for industrial establishments [7]. This determination would have to be made for the actual maintenance activity which is being carried out.

The second limitation is the lack of measured operating data. None of the sources gives a complete comparison of scaffold labour hours, permanent platform setup time, project costs, or the incidents which are associated with each arrangement. In addition, there is a concern about the independence of the information. Hatch and ZARGES are both describing solutions which are connected to their own businesses. Their publications help to explain the applications, but independent records would give much stronger support for any claim about performance.

Finally, the provisions of O. Reg. 213/91 were reviewed using the IHSA's reproduction of the regulation. The cited sections and the current wording should be confirmed through Ontario's e-Laws before they are relied on in a formal engineering assessment.

5. Conclusion

The McCormick project shows how a permanent modular platform can be used to deal with repeated maintenance access inside a hydropower station. At the same time, it also brings up the question of how the access equipment gets chosen in the first place. The scaffolding itself may be fine for the work which is being done on it. The problem is that every time it is installed again, more time is used up and the workers are also exposed to the hazards which come with putting it up and taking it down. In a station where the inspections happen often, this repeated work then ends up being a large part of the maintenance which is being done.

Of the three factors which were considered, worker safety gives the clearest reason to look into a permanent platform. Reducing the need to build and remove temporary scaffolding should reduce the exposure to those particular activities, although the size of the improvement was not measured in any of the sources which were reviewed. The time which could be saved is also worth looking into. The problem is that there is not enough published data at the moment to work out what that saving would actually be. Inspection and maintenance are also still not settled. A permanent platform avoids the repeated scaffold erection, but it then has to be kept structurally sound and serviceable for its whole intended life.

Taking all of this into consideration, selecting one system over the other based only on the available publications would not be recommended. Before doing this, a record of the current scaffold labour requirements, the frequency of maintenance access, and the expected service life and inspection needs of the proposed platform would be needed. These details would turn the comparison from a general argument in favour of permanent access into something which could actually support an engineering decision.

6. Questions for Senior Engineering Review

1. When turbine maintenance has to be repeated, is getting rid of the scaffold erection and dismantling enough on its own to justify a permanent platform from a safety point of view, or would measured labour hours and outage data normally be needed first before that decision is made?
2. Which legislation and which site requirements would then apply to the access system? In particular, how is it normally decided whether the work counts as construction or as ongoing industrial maintenance at a station like this one?
3. For a permanent platform which has been installed inside a draft tube, who is normally responsible for inspecting it over its service life (the station owner, the designer, or someone else), and how often and to what level of detail would that inspection then be expected to be done?

References

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[2] Government of Ontario, “Training for working at heights,” updated Apr. 21, 2026. ontario.ca/page/training-working-heights. Accessed Oct. 18, 2026.

[3] O. Reg. 213/91, Construction Projects, ss. 26, 26.1, 26.3, 130, 131, as reproduced in IHSA OHS Guide, ohsguide.ihsa.ca (Guardrails; Scaffolds – Erection; Scaffolds – Inspection). Accessed Oct. 18, 2026.

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